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In silico characterization of the cross-sectional turbulence of a PAN-derived carbon fiber and its effects on structural–mechanical properties

Yuri Jeon [a], Hongdeok Kim [a,*], Joonmyung Choi [a,*]

[a] School of Mechanical Engineering, Sungkyunkwan University, Suwon, 16419, Republic of Korea

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ABSTRACT

In this study, we developed an all-atom modeling framework to investigate the cross-sectional turbulent morphology within a single polyacrylonitrile (PAN)-derived carbon fiber. The modeled cross-sectional microstructure reproduced the carbonization process of ladder-structured PAN by enabling spontaneous bond formation based on the intrinsic reactivity represented by the bond-order potential. A turbulent morphology became pronounced at a hydrogen removal ratio of 0.82, corresponding to a density of 1.96 g/cm^3. Our results revealed that the development of this turbulent morphology was governed by the nonlinear growth of extended, flexible carbon clusters. The transition from linear to turbulent growth was induced by the loss of buckling strength and an increase in the flexibility of the constituent graphene layers. The proposed model also revealed that carbonization promoted the simultaneous formation of well-ordered dense lamellar regions and randomly arranged crowded regions. Notably, the incomplete bonding configurations within crowded regions caused nanopores to emerge at the dangling ends. Mechanical analysis showed that the axial and transverse moduli reached 771 and 2.4 GPa, respectively. The transverse modulus exhibited a nonlinear increase with the progression of the turbulent morphology, highlighting its critical role in the macroscopic mechanical anisotropy of carbon fiber. These findings provide theoretical guidance for the microstructural optimization and design of high-performance carbon fiber.

1. Introduction

Carbon fiber (CF) has been widely employed as a primary structural material in advanced engineering applications owing to its outstanding mechanical strength, low density, and excellent thermal stability [1–3]. CF is typically produced by pyrolyzing precursor fibers, including polyacrylonitrile (PAN) [4], polyimide [5], cellulose [6], rayon [7], and pitch [8]. Among these, PAN has emerged as the most suitable precursor for high-performance CF production owing to its superior processability and high carbon yield [9,10]. During low-temperature stabilization, linear PAN chains undergo intramolecular cyclization reactions, forming a thermally stable, non-melting ladder structure that facilitates subsequent high-temperature carbonization [11–13]. Because the gaseous byproducts released during carbonization predominantly originate from minor non-ring-forming elements, PAN-derived CF exhibits a high carbon content, exceeding 50% relative to the precursor mass [14,15]. Consequently, PAN-derived CF currently accounts for nearly 90% of the commercially produced high-strength CF worldwide [16,17].

The mechanical properties of CF, including the tensile strength and stiffness, are governed not only by the carbon content but also by microstructural features such as crystallinity, molecular orientation, and defect density [18]. A comprehensive understanding of CF microstructure is therefore essential for advancing technologies for manufacturing high-strength CF. To that end, significant experimental efforts have been devoted to characterizing nanoscale structural features. For example, high-resolution transmission electron microscopy has been extensively employed to directly measure the interlayer spacing of crystalline graphite domains at the atomic scale [19,20]. Meanwhile, wide-angle X-ray scattering has provided quantitative information regarding crystallite size and morphology [21,22]. By integrating Scherrer and Debye function analyses, wide-angle X-ray scattering can be used to estimate crystallite size distributions as well as the degree of crystallite orientation relative to the fiber axis, thereby enabling an in-depth assessment of microstructural anisotropy in CF [23]. In addition, Raman spectroscopy has been employed to gain quantitative information about defects within the crystalline domains of CF, including the sp^2–sp^3 carbon distribution [24,25]. Finally, small-angle X-ray scattering has identified nanoscale pores in the range of 30–100 nm, which degrade fiber strength

* Corresponding authors.
E-mail addresses: khd12323@skku.edu (H. Kim), joonchoi@skku.edu (J. Choi).

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[26].

Another important consideration is the cross-sectional interior microstructure that emerges during CF fabrication. The distinctive structural feature of PAN-derived CF is the cross-sectional turbulent morphology of the interior of an individual CF, which is composed of entangled turbostratic graphite [27–29]. Highly aligned PAN-derived CF exhibits an anisotropic nanostructure in which carbon atoms are covalently bonded within the graphene planes, while adjacent layers interact through van der Waals forces [30]. The fundamental building blocks of the turbulent morphology are nanoscale layered crystallites (< 4 nm) interspersed with amorphous carbon and nanopores [31]. However, owing to the extremely fine spatial scale of these features, experimental techniques face intrinsic limitations in resolving such turbulent characteristics with sufficient resolution. As a result, the mechanisms underlying the formation of turbulent structures and their influence on material properties remain only partially understood.

To overcome these limitations and probe the fine-scale features of turbulent morphology beyond experimental resolution, simulations have been employed to construct cross-sectional structures from ladder-structured PAN [32]. In this previous approach, adjacent PAN units in a ladder structure always covalently bonded whenever the unit array satisfied predefined distance and angle constraints. Although this method could clearly construct PAN-derived CF structures, the final shape of the generated cross-sectional structure is very sensitive to the distance and angle criteria, which limits the dynamic exploration of the cross-sectional structure evolution process. As a result, these approaches may not accurately represent the intrinsic mechanisms governing the development of turbulent structures and their relationship to mechanical properties.

Motivated by these challenges, this study developed an all-atom modeling framework to investigate the dynamic evolution of cross-sectional turbulent morphologies in PAN-derived CF. The framework explicitly reproduces the carbonization process of ladder-structured PAN by allowing spontaneous bond formation based on the intrinsic reactivity represented by the bond-order potential. By gradually removing hydrogen atoms that had filled the bond order, the corresponding empty spaces and reactivity dynamically influenced the structural evolution of carbon clusters. Notably, this approach successfully captured the dynamic evolution of the characteristic curvilinear and turbulent morphologies within the internal cross-section of CF. In particular, the transition from linear to turbulent growth of carbon clusters was interpreted based on the inherent flexibility of graphene. Furthermore, the impact of the increased turbulence intensity on the nanopore distribution and mechanical properties was systematically evaluated. The findings provide mechanistic insights into the structure–property relationships governing CF performance and offer theoretical guidelines for designing optimized microstructures with enhanced material properties suitable for next-generation high-performance CF systems.

2. Models and methods

2.1. Model preparation

The modeling process developed in this study directly borrowed ideas from the basic chemical process of experimentally fabricating CF. Typical processes for producing CF from PAN involve three sequential stages [33,34]: alignment, stabilization, and carbonization (Fig. 1a). During the alignment stage, the backbone of the PAN polymer is uniaxially oriented using techniques such as electrospinning and drawing. Such alignment maximizes the stacking of C≡N bonds perpendicular to the backbone chain, thereby accelerating the subsequent chemical modification rate and minimizing thermally induced structural distortion. During stabilization, exposure to temperatures of 500–600 K converts the unit molecular structure into a closed loop, i.e., cyclization, while removing hydrogen atoms within the PAN chain, transforming PAN into an insoluble ladder structure. Finally, PAN is carbonized by applying very high temperatures of 1000–3000 K, during which the ladder-structured units covalently interconnect with their neighbors, transforming PAN into a complex graphene-based two-dimensional structure.

Overall, the process of converting a ladder structure into a two-dimensional carbon planar one involves significant nonlinear microstructural changes that influence the curved and turbulent morphology of the final CF interior. In particular, the evolution of the cross-sectional structure, which is the key focus of this study, is primarily determined during the growth of the ladder-structured PAN, i.e., the carbonization stage. Therefore, we aimed to design the microstructures of well-

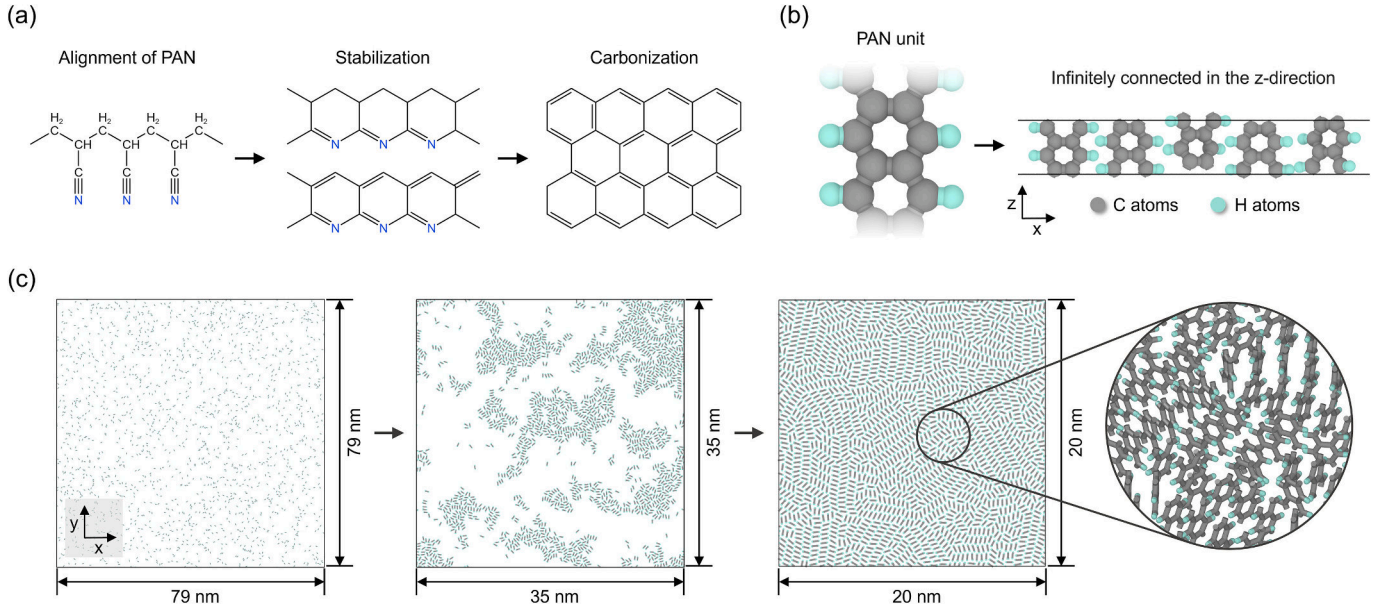


Fig. 1. All-atomic modeling of the cross-sectional morphology of PAN-derived CF. (a) Typical process for preparing PAN-derived CF and the associated molecular structural changes. (b) Ladder-structured PAN units for observing the cross-sectional structural evolution of CF. (c) Preparation of a microstructure consisting of highly oriented ladder-structured PAN.

organized ladder-structured PAN and then describe their growth process within an all-atom framework. As a modeling strategy, the adoption of an idealized C/H-only structural model assumes that nitrogen and oxygen have already been released from the system. The turbostratic structure is intrinsically associated with the dynamic evolution of flexibility arising from the growth of carbon clusters. From this perspective, the flexibility of aromatic clusters is primarily governed by the intrinsic properties of the carbon-dominant bulk structure. Therefore, nitrogen and oxygen-containing groups primarily located at the edges are excluded from this mechanism.

The detailed procedure for modeling the PAN microstructure was as follows: First, a ladder-structured PAN polymer was prepared in a molecular dynamics (MD) simulation environment (Fig. 1b). The ladder-structured PAN consisted of eight carbon atoms arranged in a hexagonal configuration with hydrogen termination. To describe the high molecular weights of polymers, the z-axis of the unit cell was set to be exactly twice the height of a benzene ring (4.84 Å), and ladder-structured PAN unit were infinitely connected beyond the z-boundary. The xy-plane corresponding to the cross-sectional direction of the PAN fibers was sufficiently large to accurately capture the microstructural variations. A total of 1765 ladder units were randomly distributed within a periodic unit cell of 79 nm × 79 nm (Fig. 1c). The unit cell was prepared with a low initial density of 0.1 g/cm^3. This low density minimized atomic overlap during random insertion and avoided excessively high repulsive energy. Furthermore, it prevented non-physical phenomena such as ring catenation in cyclic molecules such as ladder-like PAN. This low-density initial microstructure was constructed as a computational strategy to avoid numerical problems and stabilize the design. This approach has been widely used, particularly in the molecular-scale modeling of polymers and polymer nanocomposites [35,36]. The total energy of the system was minimized using a conjugate gradient algorithm with a convergence criterion of 0.001 kcal/mol. Subsequently, the system was gradually densified for 2.5 ns under an isothermal–isobaric ensemble (NPT; controlled by the number of atoms N, pressure P, and temperature T) ensemble at a temperature of 100 K, applying external pressures of 500 atm in the x- and y-directions and 1 atm along the z-direction. Through this process, the system reached a densified state with a final density of approximately 1.55 g/cm3. This structure was used as the starting configuration to simulate carbonization. All interatomic interactions were described using the AIREBO-Morse potential [37].

2.2. Carbonization protocol

The ladder-structured PAN was carbonized by leveraging the inherent reactivity represented by the AIREBO-Morse potential. When the benzene rings contain unsaturated bonding sites, the AIREBO-Morse potential induces spontaneous covalent bond formation between adjacent rings separated by less than 3.69 Å [37] (Fig. 2a). This spontaneous reactivity was exploited to accurately capture the cross-sectional morphology changes of the CF. Specifically, under an NPT ensemble identical to that of the densification process, 10% of the hydrogen atoms present in the simulation cell were removed from random locations. Covalent bonds spontaneously formed between unsaturated PAN units when the inter-unit distance was within the reactive trapping range. Meanwhile, the formation of covalent bonds rapidly increased the kinetic energy, resulting in a transient temperature spike. Therefore, we introduced a sufficient relaxation time of 0.25 ns during each removal cycle to ensure that the entire system equilibrated at 100 K. The adoption of the low-temperature condition of 100 K is explained in detail in Supplementary note 1 and Fig. S1. The dehydrogenation cycle was repeated 40 times, corresponding to a hydrogen removal ratio of 0.96. Subsequently, all remaining hydrogen atoms in the unit cell were completely removed owing to their low content. Further stabilization was performed for 0.5 ns under the NPT ensemble. During this process, the hydrogen removal ratio increased nonlinearly with repeated dehydrogenation cycles (Fig. 2b). After 30 dehydrogenation cycles, the carbon content of the model reached 93%, which was consistent with the experimentally reported carbon content range (93–98%) for typical CF [38].

The size distribution of the carbon clusters was analyzed for each hydrogen removal ratio (Fig. 2c). The cluster size was defined as the total number of PAN units within a covalently bonded sheet. A significant number of hydrogen atoms were rapidly removed during the early removal cycles. However, unsaturated rings rarely encountered each other, delaying the growth of the cross-sectional structure (Fig. 2d).

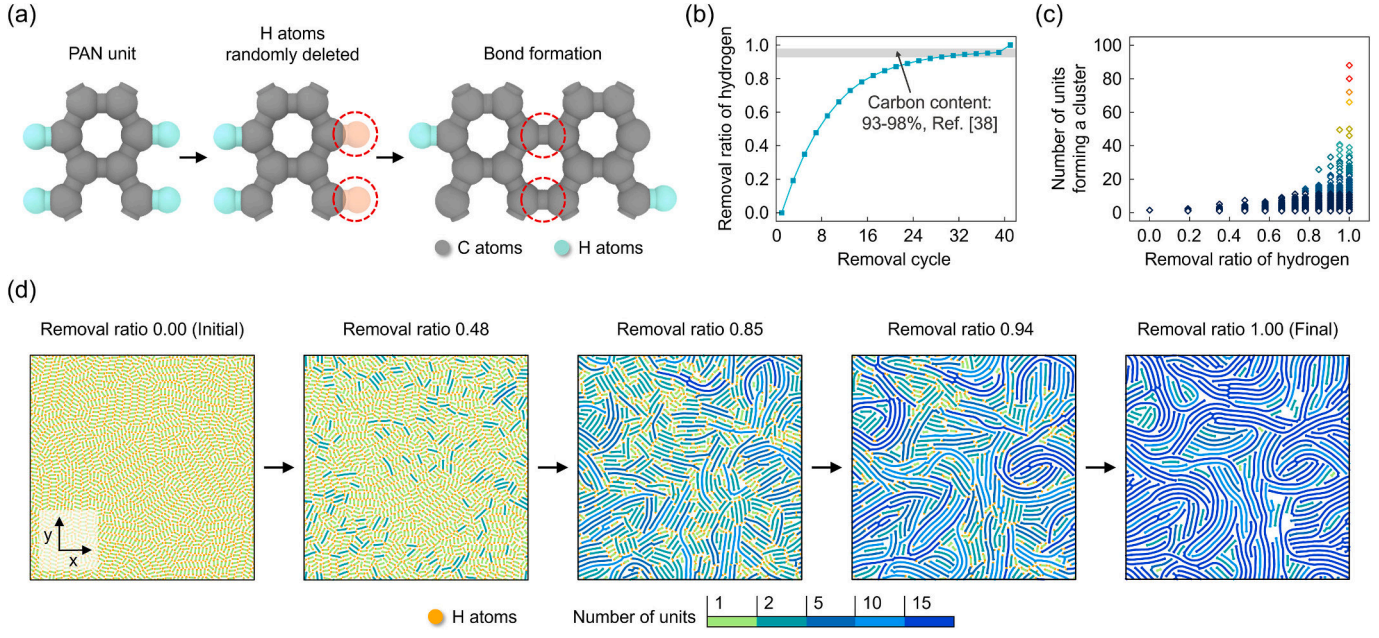


Fig. 2. Simulation of the growth of CF cross-sectional structures. (a) Schematic of the covalent bond formation process between ladder-structured PAN units. (b) Fraction of hydrogen atoms removed as a function of the dehydrogenation cycle. (c) Carbon cluster size distribution versus the hydrogen removal ratio. (d) Visualization of the progressive evolution of the CF cross-sectional morphology.

Substantial cross-sectional structural growth was observed predominantly at removal ratios of >0.5, with larger clusters having sizes exceeding 15 units beginning to emerge. Owing to geometrical constraints, these larger clusters significantly lost their linearity, forming the characteristic turbulent morphology typically observed in CF, as discussed in detail in Section 3.2.

In addition, the ladder-structured PAN microstructure was modeled and visualized using the commercial MD software Materials Studio 2020 (Dassault Systèmes, France) and an open visualization tool [39]. Carbonization and property evaluations under a thermodynamic ensemble were performed using a large-scale atomic/molecular massively parallel simulator package (Sandia Laboratory, USA [40]). The time integration was based on the velocity Verlet algorithm with a time step of 0.5 fs.

3. Results and discussion

3.1. Model validation

The modeling procedures and methodology were validated by analyzing the microstructure. First, the density of the model was examined as a function of the hydrogen removal ratio (Fig. 3a). The model density increased progressively with the removal of hydrogen atoms, especially in the later stages, when the formation of C—C bonds became dominant. The density of the as-prepared microstructure, i.e., cyclized PAN fiber, was 1.55 g/cm^3, which was higher than the experimental value of 1.17 to 1.22 g/cm^3 for typical PAN-based fiber [41]. Such high-density conditions are understood as an ideal regime achievable when the design assumes the complete orientation and rich cyclization of the PAN chains. After the carbonization simulation was completed and all hydrogen atoms were removed, the final density of the microstructure reached 2.22 g/cm^3. The value is close to the theoretical density of 2.25 g/cm^3 [42] of graphite crystals, which means that a completely ideal carbon microstructure formed, except for structural defects that arise spontaneously from the order in which all carbon atoms bond to each other. The cross-sectional structures of the CF and graphite models are compared in Supplementary note 2 and Fig. S2. In particular, the model excluded the formation of large-scale pores that are commonly observed in experimentally produced PAN-derived CF. The nitrile groups of PAN precursors participate in a well-known sequential cyclization, which promotes the formation of ladder-like structures. At later stages, the evolution of volatile species, such as NH_3 and HCN, forms elongated pores [43], and the high drawing ratio applied during spinning and stabilization aligns these pores along the fiber axis. Previous studies have shown that typical CF has an experimental density of 1.75 to 1.95 g/cm^3 [44]. During actual CF manufacturing, the formation of micropores, structural defects, and amorphous carbon lowers the overall average density of the structure. By contrast, the carbon network density of the microstructure considered in this simulation was increased purely through dehydrogenation. In this system, the kinetic capture of generated gases was completely excluded; therefore, no large voids remained within the structure, resulting in a higher final density than that of experimental systems that include such voids.

The relative proportion of sp^2–sp^3 carbons was monitored during carbonization (Fig. 3b). As hydrogen atoms were removed at a constant rate, the number of C—C bonds increased exponentially. In other words, in the early hydrogen-rich PAN chain environment, the formation of new C—C bonds and the resulting evolution of the cross-sectional morphology were not active. These geometric constraints and the resulting insensitivity to changes in the initial shape were in good agreement with previously observed characteristics of CF manufactured through the carbonization of existing PAN chains (Fig. 2c) [45]. Abundant C—C bonds formed only after the hydrogen removal rate exceeded 0.4. Notably, the analysis of the sp^2–sp^3 ratio showed that the fraction of sp^3 carbon associated with amorphous carbon was negligible throughout the carbonization process. This suggests that the simulation environment parameters should be set such that the overall morphology converges to an ideal turbulent structure unless the user intentionally introduces amorphous structures or defects.

To confirm the crystallographic validity of the completed CF model, an X-ray diffraction (XRD) profile was computed (Fig. 3c). The first diffraction peak appeared at 26.8°, corresponding to the interlayer spacing of graphite (d_{002}). The peaks at 44.0° and 78.9° corresponded to the (100) and (110) planes of the graphite crystal structure, respectively. Overall, this XRD profile was highly consistent with distributions previously calculated for fully carbonized fibers [46], suggesting that the simulated microstructure successfully reproduced the structural features of CF.

3.2. Formation process of cross-sectional turbulent structure

The evolution of the turbulent morphology along the cross-sectional direction of the CF was investigated throughout the hydrogen removal process (Fig. 4). The orientation angle was defined by constructing vectors between two adjacent PAN units (Fig. S3). First, the centroid of each PAN unit was calculated from the atomic positions of the carbon backbone. Then, vectors passing through neighboring centroids were calculated independently. The orientation angle between two PAN units was defined by measuring the angle between adjacent vectors. The PAN unit exhibited values close to 180° in the linear configuration (light blue) and 160° in the curved configuration (purple). This orientation angle can be used to quantify the degree to which the turbulent morphology developed along the cross-sectional direction of the CF. This analysis suggested two notable findings. First, in the early stages of carbonization (hydrogen removal ratio ranging from 0 to 0.73), carbon clusters predominantly retained linear morphologies. Although C—C bonds rapidly formed, and adjacent PAN units coalesced beyond a removal ratio of 0.5 (Fig. 3b), the development of turbulent topology remained negligible at this stage. Second, pronounced turbulence emerged only after sufficient cluster growth, specifically when the hydrogen removal ratio exceeded approximately 0.85. These findings suggested that the evolution to turbulent morphology followed a nonlinear progression with respect to the C—C network development. In

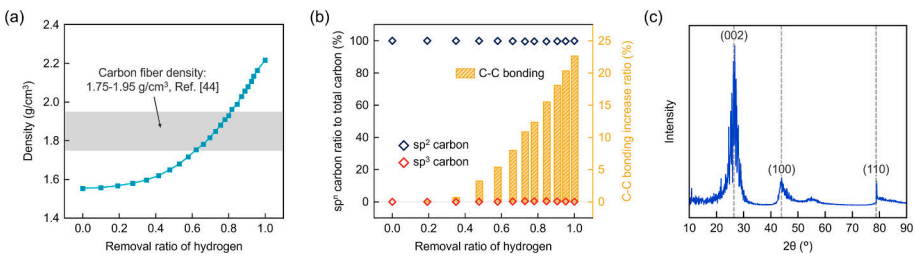


Fig. 3. Structural characterization of PAN-derived CF during the carbonization process. (a) Variation in the mass density with hydrogen removal ratio. The shaded region indicates the experimental range of typical CF (1.75–1.95 g/cm^3). (b) Changes in the carbon hybridization states (sp^2 and sp^3) and C—C bonding ratio as a function of hydrogen removal. (c) XRD pattern of the completed CF microstructure.

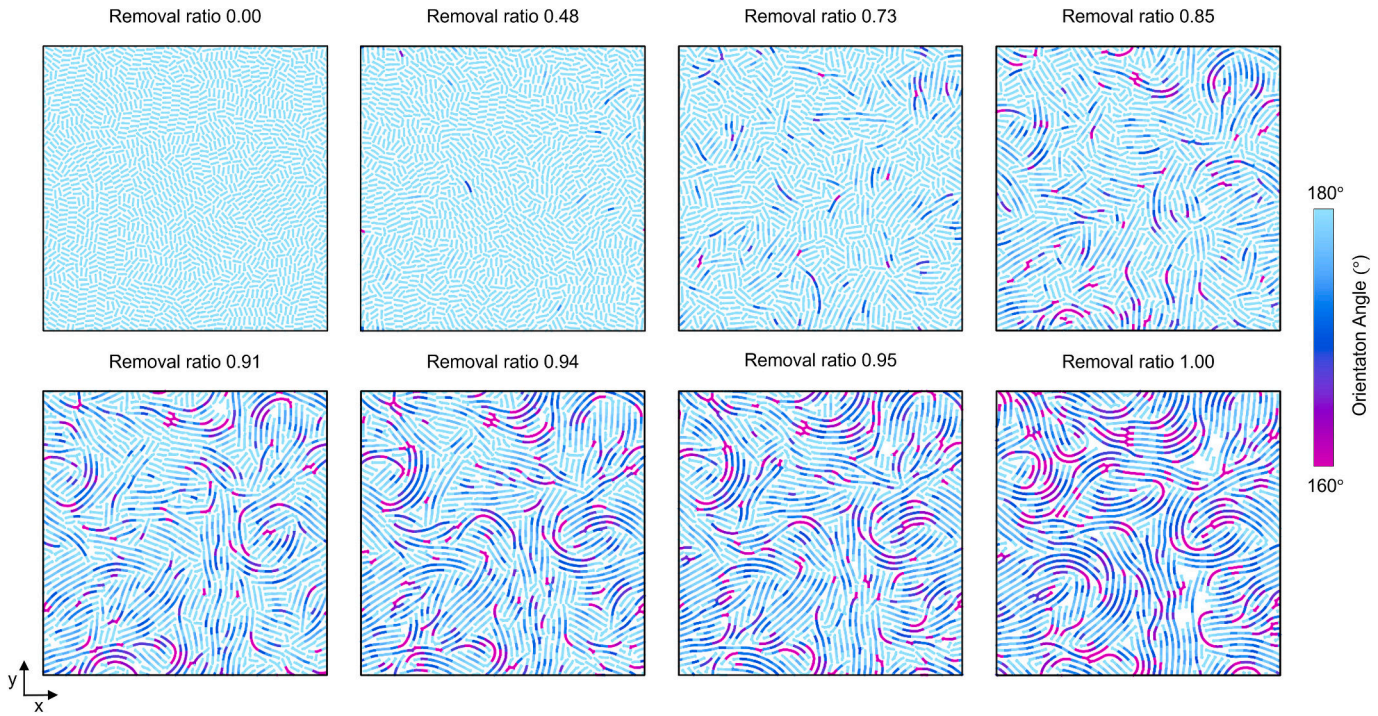


Fig. 4. Evolution of turbulent-like cross-sectional structures in CF as a function of the hydrogen removal ratio. The configurations are color-mapped based on the orientation angle.

other words, the onset of turbulent structural organization was governed not only by the accumulation of covalent bonds but also by the geometric constraints and the aggregation of carbon clusters.

The molecular-level origin of the turbulent morphology was further elucidated based on the structural flexibility of the graphene sheets. To evaluate the bending resistance according to the cluster size, a single-sheet graphene model was constructed, and uniaxial buckling tests were performed (Fig. 5a). The number of PAN units constituting graphene varied from 2 to 20. The length of the simulation box was set to the equilibrium distance of each sheet (i.e., along the x- and z-axes), and

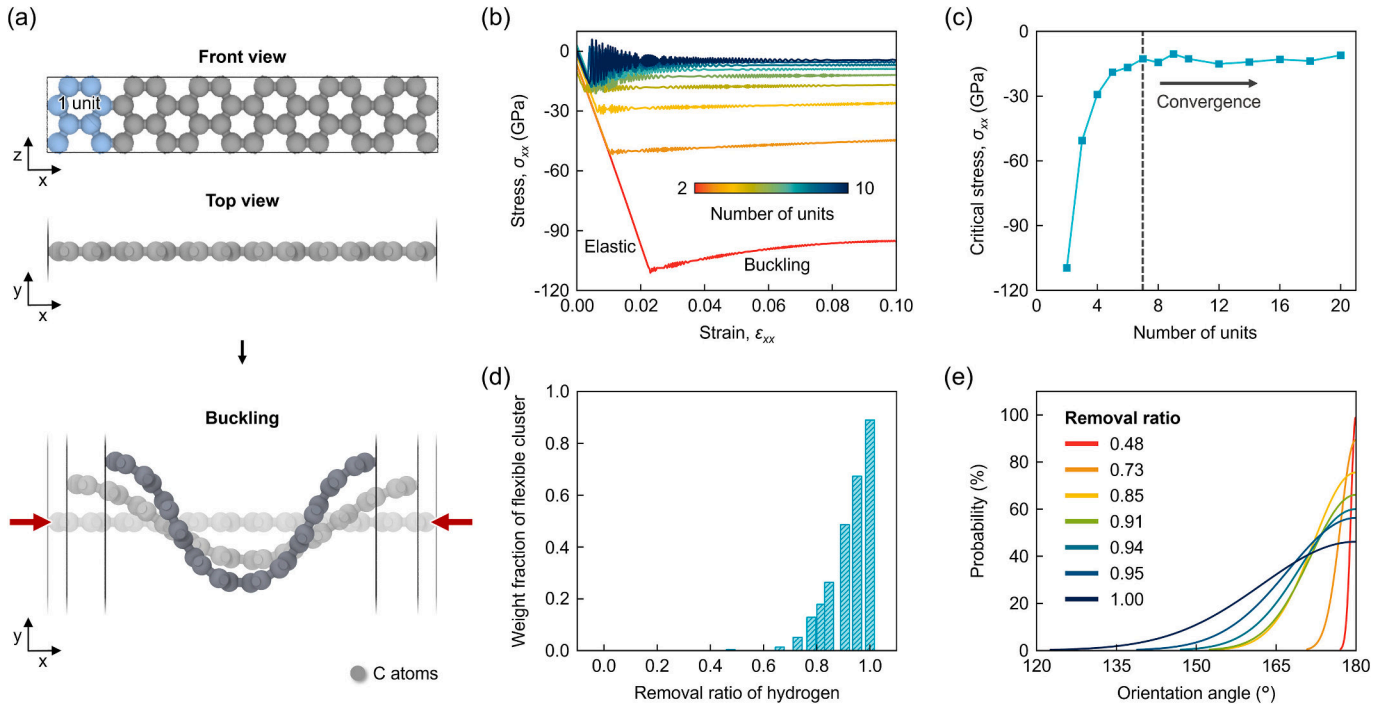


Fig. 5. Buckling simulations of single-sheet graphene. (a) Atomistic configuration of the single-sheet graphene model. The model began with two connected PAN-derived sheets, and the number of constituent units gradually increased from 2 to 20 units. (b) Stress–strain curves of graphene under uniaxial compression. (c) Variation in the critical buckling stress with the number of constituent PAN units. (d) Weight fraction of flexible carbon clusters (i.e., constituent units >7) during the removal of hydrogen from the CF model. (e) Histogram of the orientation-angle distribution according to the hydrogen removal ratio.

sufficient clearance was provided in the normal direction (i.e., the y-axis) to avoid interference across the periodic boundary during buckling. Each constructed model was equilibrated for 100 ps under a canonical ensemble (NVT; controlled by the number of atoms N, volume V, and temperature T) at 0.1 K. Subsequently, uniaxial compression was applied along the x-direction with a constant strain rate of 5×10^8 s^{-1}. Deformation continued until a compressive strain of 30% was reached, and the stress tensor along the loading direction was evaluated based on the virial theorem [47].

Under compressive deformation, the graphene sheet preserved its planar configuration. The stress–strain profile also showed a linear decrease corresponding to elastic compression (Fig. 5b). However, above a critical threshold, graphene buckled significantly in its normal direction, dissipating the internally accumulated stress. Importantly, the critical buckling stress was significantly influenced by the number of ladder-structured units constituting the graphene. Increasing the graphene length promoted buckling even at low stress levels, enhancing its adaptability to bending and curved conformations. The critical stress was evaluated as a function of the number of ladder-structured units (Fig. 5c), and it was found to increase logarithmically with the graphene length and to converge beyond seven units. In conclusion, the bending resistance of the carbon clusters significantly deteriorated as it exceeded the critical size during carbonization, thereby ensuring structural flexibility.

To correlate the structural flexibility of the carbon clusters with the evolution of the turbulent morphology, we measured the proportion of flexible clusters (i.e., units >7) during the carbonization process (Fig. 5d). Flexible clusters were absent in the early carbonization stage and began to appear only when the hydrogen removal ratio exceeded 0.8. This onset coincided very well with the point at which low orientation angles (~160°) began to appear in the atomic snapshots (Fig. 4). Such structural changes could be further quantified based on the orientation angle distribution (Fig. 5e). The angular distribution, which was initially biased toward 180°, began to rapidly disperse to lower values near the point at which the hydrogen removal rate exceeded 0.85.

Overall, these results demonstrated that the turbulent morphology observed in the CF cross-section was governed by the flexibility of the carbon clusters, which was therefore influenced by the absolute length of the spatially confined sheets. The early immature clusters were characterized by high bending stiffness owing to their short length; thus, they tended to individually maintain a linear configuration. Turbulent structures arose only after extensive hydrogen removal, thereby forming sufficiently large, flexible clusters with lower bending stiffness. Admittedly, such a simplified model did not fully account for the complexity of actual CF microstructures, including inter-cluster interactions such as van der Waals forces and hydrostatic pressure under condensed-state conditions. Nevertheless, examining a single-cluster model is highly significant as it serves as a fundamental starting point to isolate the intrinsic mechanical properties of the structural building blocks, independent of the stochastic nature of crosslinks or the local packing environments in a bulk system. From this perspective, the present results clearly indicated that individual carbon clusters possessed a lower-bound threshold at which structural flexibility began to emerge. Thus, the occurrence of turbulence during carbonization was closely related to the growth of carbon clusters and the resulting nonlinear changes in structural flexibility.

3.3. *Evolution of nanoscale cavities in CF cross-sections during carbonization*

The evolution of nanopores in the CF cross-section was analyzed throughout the carbonization process (Fig. 6a). For each atom within the microstructure, a Voronoi cell was constructed based on its spatial position relative to neighboring atoms. The cavity radius was then defined as the distance from that atom to the furthest vertex of the Voronoi cell; thus, this parameter quantitatively measured the local free volume size of each constituent atom. In the low-density microstructure corresponding to the initial carbonization stage (hydrogen removal ratio:

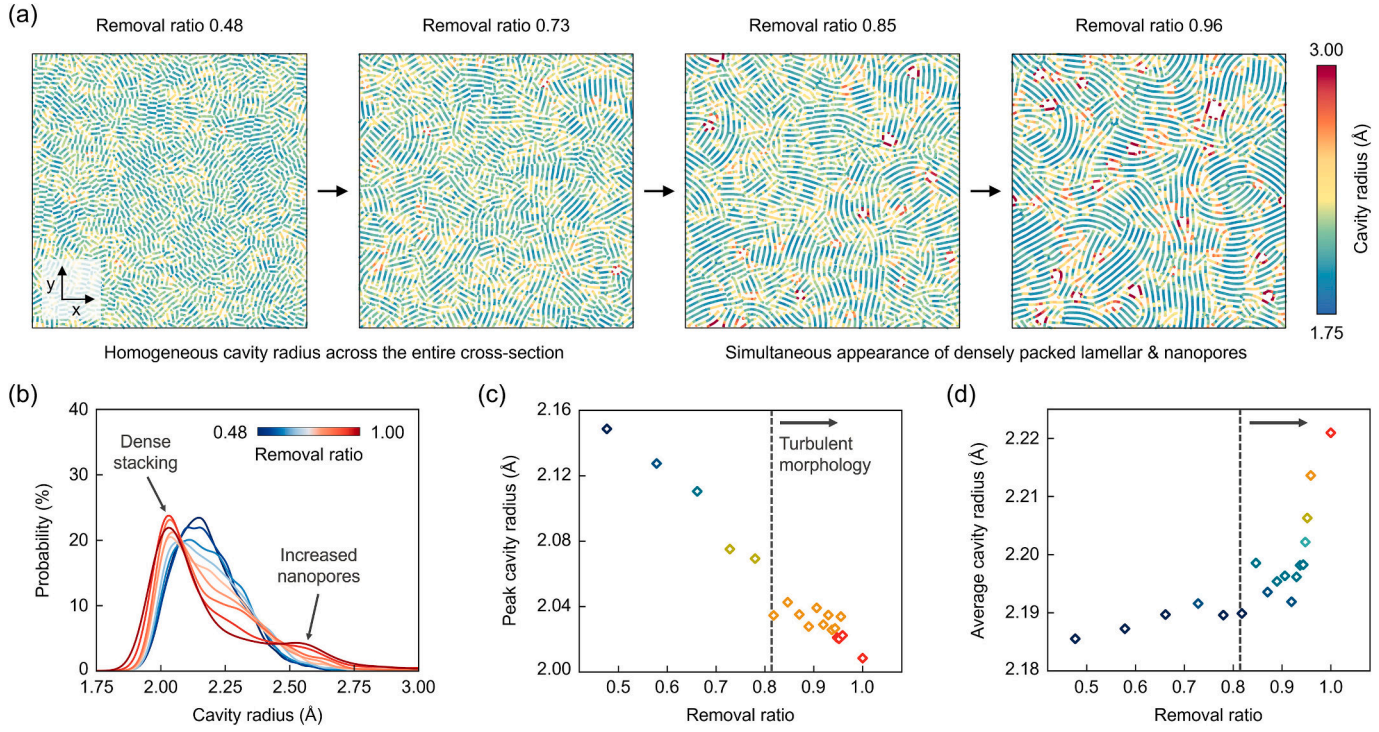


Fig. 6. Evolution of nanopores in a CF cross-section during carbonization. (a) Atomistic visualization of lamellar and crowded areas at various stages of carbonization. The constituent atoms are colored according to their cavity radii. (b) Histogram of the cavity radius distribution as a function of the hydrogen removal ratio. This profile was fitted using a kernel density estimation method. (c) Variations in the peak cavity radius and (d) atom-averaged cavity radius as functions of the hydrogen removal ratio.

0.48–0.73), the cavity radius was distributed homogeneously across the entire cross-section. Owing to their high tolerance to structural reorganization based on their high molecular mobility, nanopores evolved negligibly at this stage. However, as the hydrogen removal ratio gradually increased (hydrogen removal ratio: 0.82–1.00), the cross-section structure was divided into regions with two distinct structures: well-ordered dense lamellar domains and randomly arranged crowded regions. PAN molecules, which initially could arrange themselves freely, gradually became less mobile and more rigid as carbonization progressed. This rigidity acted as a seed, which further inhibited the arrangement of adjacent PAN molecules. As this process occurred simultaneously in multiple locations within a given space, it created localized congested regions composed of disordered and randomly arranged carbon clusters. This structural congestion limited efficient atomic packing, thus promoting nanopore formation. In the packed lamellae, carbon atoms formed a regularly layered structure, resulting in a high-density region characterized by a small cavity radius (~2 Å). Conversely, in the crowded areas, the abundance of CH-terminal ends and imperfect bonding led to the formation of expanded nanopores, increasing the cavity radius (~2.5 Å). Fig. S4 schematically illustrates how the overall carbonization process ultimately forms these expanded nanopores through sequential structural changes from individual PAN units into a complex carbon network.

To quantitatively characterize the nanopore features, the distribution of cavity radii was obtained according to the hydrogen removal ratio (Fig. 6b). The profiles of the low hydrogen removal ratio models exhibited a unimodal distribution with an intermediate peak cavity radius (~2.2 Å). As carbonization proceeded, the peak position continuously shifted toward a smaller cavity radius, while the probability of a cavity radius of >2.5 Å also increased noticeably. This behavior indicates that carbonization simultaneously formed condensed carbon and nanoporous domains. The variations in the histogram peak position and average cavity radius were further measured (Fig. 6c and d). The peak cavity radius decreased monotonically with the increasing hydrogen removal ratio and converged at 0.82, at which point turbulent structures developed. Furthermore, despite the growth of packed areas, the average cavity radius increased exponentially with the growth of the turbulent structures. Nanopores emerged as a structural consequence of this evolving turbulent morphology, and their formation was intrinsically linked to the carbonization pathway.

These insights suggest that controlling the development of the turbulent morphology during carbonization may provide an effective strategy for tailoring the internal porosity of CF. The formation of crowded regions and the associated nanopores can be mitigated by directing the carbon microstructure toward a more ordered configuration. One promising approach would be to incorporate carbon nanomaterials into CF precursors, e.g., graphene- or carbon-nanotube-related nanomaterials. Indeed, previous studies showed that adding graphene to PAN-based precursors promoted the in-plane orientation of graphitic domains and suppressed pore and defect formation, thus forming a less porous cross-sectional structure [48]. By analogy, introducing low-dimensional carbon nanostructures is expected to suppress the formation of crowded regions, thereby limiting the nucleation of the large nanopores that contribute disproportionately to the average cavity radius.

3.4. *Influence of turbulent morphology on the mechanical properties of CF*

The effect of the turbulent morphology on the mechanical properties of the CF cross-section was investigated by performing uniaxial tensile tests on models with different densities. The CF microstructures corresponding to different stages of carbonization were extracted at 0.5 ns intervals during the hydrogen removal simulation. Each model was further equilibrated for 25 ps under an NVT ensemble at 100 K. Subsequently, uniaxial tensile deformation was applied along each axis of the unit cell under an NPT ensemble at 100 K and 1 bar with a constant strain rate of $5 \times 10^8\ s^{-1}$. In addition, note that the strain rate considered was sufficiently low to capture the nearly quasi-static deformation behavior of molecular structures in MD frameworks [49–51]. The CF model exhibited negligible sensitivity to the strain rate because its viscoelastic response was limited (Fig. S5). During uniaxial loading, the stress applied to the CF constituent atoms was calculated using the virial theorem [47]. For statistical reliability, each tensile test was repeated five times along each axis, and the results were averaged.

The axial (Fig. 7a) and transverse elastic moduli (Fig. 7b) were determined from the linear slopes of the stress–strain curves up to a strain of 0.03. The axial modulus increased linearly with the CF density, which is consistent with the experimentally reported trends for CF [52,53]. The axial modulus was 771 GPa at the final density of 2.22 g/cm^3, thus increasing by approximately 37% relative to 561 GPa at the initial density of 1.65 g/cm^3. During densification, the load-inefficient dangling ends were progressively eliminated, and the continuities of the graphitic domains along the CF cross-section were enhanced, enabling efficient load transfer in the axial direction. In this regard, the axial modulus is compared with that of an ideally stacked graphite model in Supplementary note 3 and Figs. S6 and S7. The moduli obtained in this study were higher than the typical experimental values of 200–300 GPa [54]. This could be explained by the idealized model, in which structural defects and large-scale pores were minimal, and all PAN units were perfectly aligned parallel to the fiber axis. Such an overestimation is therefore reasonable and is commonly observed in atomistic simulations [55]. Nevertheless, additional simulations were performed to improve the physical validity of the model by introducing structural defects and controlling orientation dispersion. For additional results, see Supplementary note 4 and Figs. S8 and S9. By contrast, markedly different behavior was observed in the transverse direction. The transverse modulus increased nonlinearly with the CF density and rose sharply beyond approximately 1.96 g/cm^3, at which point the turbulent morphology became prominent. At the density of 1.64 g/cm^3, the transverse modulus was 0.94 GPa, which increased by ~57% to 1.48 GPa at the onset density of the turbulent morphology (1.96 g/cm^3). At the final density of 2.22 g/cm^3, the modulus reached 2.39 GPa, an increase of approximately 154% relative to that of the initial state. This value also corresponds to an additional increase of ~97% relative to that of the turbulence onset density (1.96 g/cm^3). This trend is consistent with the decrease ratio of the average orientation angle (Fig. S10). Moreover, this behavior indicates that the transverse modulus was strongly governed by the combined effects of densification and the development of cross-sectional turbulent structures, suggesting that turbulence-induced structural interlocking played a critical role in enhancing the transverse load-bearing capability.

To further investigate the influence of the turbulent morphology on the transverse mechanical behavior, the evolution of shear strain in the CF cross-sections during tensile deformation was analyzed. Two representative microstructures were selected to characterize the deformation mode as carbonization progressed: one at the initial carbonization stage (density = 1.85 g/cm^3) and the other after the turbulent morphology formed (density = 2.15 g/cm^3). As the tensile strains of 0.005, 0.01, and 0.03 were applied to the microstructures, the internal shear strain distributions based on the Green–Lagrangian strain tensor were monitored [56] (Fig. 8a). The results clearly indicate that the deformation mode was significantly affected by the turbulent morphology. For the 1.85 g/cm^3 model, regions of low (blue) and high (red) shear strain were homogeneously distributed throughout the cross-section. Owing to the weakly interlocked architecture, the transverse strain induced collective and clustered behavior in the ladder-structured PAN, thereby lowering the load transfer efficiency. In contrast, the 2.15 g/cm^3 model exhibited a noticeable region of low shear strain. In other words, the physical interlocking induced by the turbulent morphology effectively prevented slippage between the carbon layers, thereby providing high resistance to external deformation.

To quantitatively evaluate these observations, histograms of the

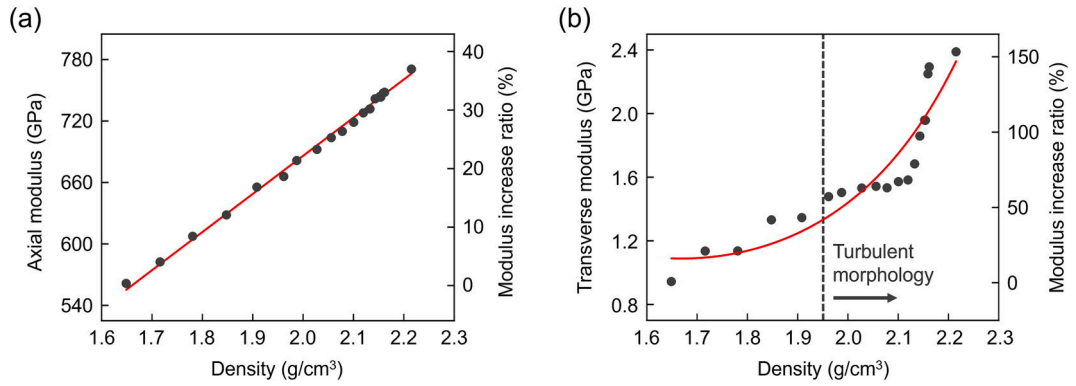


Fig. 7. Elastic moduli of the modeled CF obtained from uniaxial tensile tests. (a) Axial and (b) transverse moduli (left) calculated from tensile deformations applied along the z-direction and the averaged x- and y-directions, respectively, along with the corresponding increases in modulus (right). The solid red line represents linear fitting for the axial modulus exponential fitting for the transverse modulus.

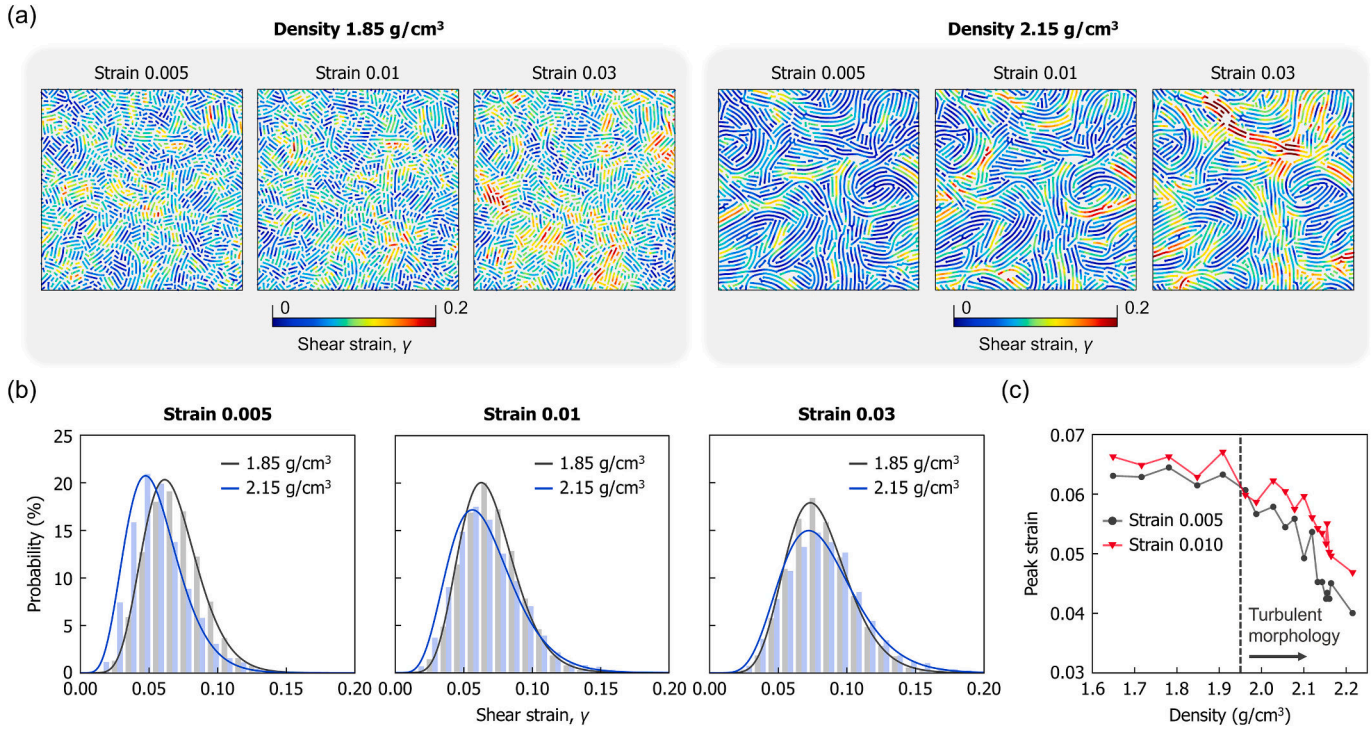


Fig. 8. Strain analysis of CF cross-sectional structures during transverse tensile deformation. (a) Snapshots of internal shear strain distributions in CF cross-sections at transverse strains of 0.005, 0.01, and 0.03. The models were obtained at the early stage of carbonization (density = 1.85 g/cm^3, left) and after the turbulent morphology formed (density = 2.15 g/cm^3, right). (b) Histograms of shear strain at transverse strains of 0.005, 0.01, and 0.03. (c) Variation in the peak shear strain as a function of the CF density at tensile strains of 0.005 and 0.010.

shear strain distributions were analyzed (Fig. 8b). The peak in each histogram represents the most probable local shear deformation state. In the early stage of tensile deformation (strain = 0.005), the distribution of the 2.15 g/cm^3 model was clearly shifted toward lower shear strain values than that of the 1.85 g/cm^3 model. This shift indicated that a larger fraction of the structure experienced minimal local shear deformation. Such behavior originated from the interlocked configuration of carbon layers in the turbulent morphology, which prevented interlayer slippage, thus enhancing the load-bearing capacity. As the applied tensile strain increased, the evolution of the distribution differed significantly between the two models. In the 1.85 g/cm^3 structure, the peak position remained nearly unchanged, which suggests that the model continued to accommodate deformation through localized shear without substantially changing the load transfer mechanism. By contrast, the peak of the 2.15 g/cm^3 structure progressively shifted toward higher shear strain. This trend indicates that although the interlocked structure initially resisted deformation, accumulated strain eventually led to partial slippage within the constrained network. The variation in the peak strain was further examined throughout the carbonization process (Fig. 8c). The peak strain remained high and almost constant for densities below approximately 1.96 g/cm^3, i.e., before the turbulent morphology formed; however, once turbulent morphology became prominent, the peak strain decreased sharply with increasing density. This result indicates that the development of turbulent morphology significantly enhanced the resistance to the initial tensile deformation by improving transverse-load-bearing capability.

Overall, the proposed modeling protocol systematically elucidated the influence of cross-sectional turbulent microstructures in CF on its mechanical and structural properties. The turbulent morphology, characterized by misaligned and distorted carbon layers, promoted

interlocking between adjacent graphitic layers while facilitating the formation of nanopores as the microstructure evolved. These dual effects give rise to a critical trade-off in property optimization. First, when the axial properties dominate the performance requirements, mitigating the turbulent structure can enhance the graphite alignment. The longitudinal stiffness and strength of the overall microstructure are also improved. Second, reinforcing the turbulent morphology is theoretically advantageous in applications where CF must withstand transverse impacts or compressive loads. The multiple turbulent morphologies formed within the cross-sectional structure enhanced interlayer interlocking, which made internal load transfer more efficient during external deformation. In parallel with recent advances incorporating nanoadditives, such as graphene and carbon nanotubes, to tailor CF microstructures, this work provided a mechanistic framework that clarified how the controlled modulation of turbulent cross-sectional architecture governs macroscopic properties. By explicitly defining the structure–property relationships associated with the turbulent morphology, these findings offer actionable design guidelines for the rational engineering of high-performance CF.

4. Conclusions

In this study, the formation of a cross-sectional turbulent morphology in PAN-derived CF and its influence on the structural and mechanical properties of the CF were systematically investigated. Ladder-structured PAN was progressively dehydrogenated within an atomistic simulation framework to describe the carbonization process. The proposed framework successfully captured the dynamic growth of carbon clusters and the emergence of a characteristic cross-sectional turbulent morphology.

Our results demonstrated that the development of the turbulent morphology was governed by nonlinear changes in the structural flexibility of the growing carbon clusters. In the early stages of carbonization, small, immature clusters with high bending stiffness primarily maintained a linear configuration; a distinct turbulent morphology emerged only after sufficient hydrogen removal, leading to the formation of extended, flexible carbon clusters. Furthermore, nanoscale cavity analysis revealed that carbonization promoted the simultaneous formation of a well-ordered, dense lamellae and a randomly arranged, crowded region. Notably, the crowded areas exhibited enlarged nanopores owing to imperfect bonding and an abundance of CH terminal groups, highlighting the important relationship between the turbulent morphology and internal porosity. Mechanical analysis further demonstrated that turbulent morphology was decisive in determining the transverse mechanical behavior of CF. As the turbulence decreased, the overall CF density increased, which linearly increased the axial modulus. In contrast, the transverse modulus increased considerably only after a turbulent morphology formed. The turbulent morphology induced strong interlocking between layers, effectively suppressing interlayer sliding between carbon sheets and improving transverse load-bearing capacity. Therefore, control of the cross-sectional turbulent structures is key to determining the mechanical anisotropy of the finished CF.

Overall, these findings provide a mechanistic understanding of the structure–property relationships associated with turbulent nanostructures in CF cross-sections. However, its direct applicability to industrial systems is limited as the present model does not fully capture the diversity of defects and heterogeneities in real CF. Nevertheless, a key contribution lies in establishing a well-controlled baseline that isolates the intrinsic effect of turbulent morphology on mechanical behavior. In line with recent studies focusing on microstructural optimization, this study can serve as a rational design guideline for high-performance CF. An extended simulation framework is currently under development to better represent industrial CF by incorporating controlled defect populations, heteroatom doping, and realistic orientation distributions. Furthermore, the established theoretical approach will be expanded in the future to include nanoadditive-driven cross-sectional modulation and longitudinal nanostructures, facilitating the sophisticated design of CF.

CRediT authorship contribution statement

Yuri Jeon: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Hongdeok Kim:** Writing – review & editing, Supervision, Conceptualization. **Joonmyung Choi:** Writing – review & editing, Supervision, Software, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could influence the work reported in this study.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2026.177773.

Data availability

Data will be made available on request.

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